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Counterparty Risk: Implications for Network Linkages and Asset Prices

Contents

1 Big picture

- **Question.** What does trade credit reveal about firms' counterparty risk, production-network links, and expected stock returns?
- **Main empirical object.** The key firm-level variable is receivables-to-sales, denoted R/S. It measures how much credit a supplier extends to customers relative to sales.
- **Fact 1: asset pricing.** Firms that extend *less* trade credit earn higher expected returns. The value-weighted low-minus-high R/Sspread is about **0.636% per month**, or roughly **7.6% per year**. The authors call this the **counterparty premium**.
- **Fact 2: network durability.** Firms that extend more trade credit maintain longer supplier-customer links. A one-standard-deviation increase in R/Spredicts about **four additional months** of future link duration and a lower probability that links break.
- **Core mechanism.** Trade credit is a way for suppliers to insure valuable customers and keep them attached. Losing a customer forces costly search and rematching, and these search costs are systematic: they are high in bad states.
- **Why low R/Sfirms are risky.** Low R/Sfirms are often matched with lower-quality customers and are more likely to lose customers. They therefore face more exposure to aggregate customer-search-cost shocks.
- **Why high R/Sfirms are safer.** High R/Sfirms extend credit to higher-quality customers worth retaining. Trade credit lowers customer departure/default probabilities and lengthens relationships, reducing exposure to costly search.
- **Aggregate prediction.** Aggregate trade credit proxies for customer-search costs. When search costs rise, firms want to extend more credit to retain customers, so aggregate R/Srises.
- **Asset-pricing test.** Innovations to aggregate R/Sexplain the counterparty premium, while standard factor models and many macro factors do not.
- **Model contribution.** The paper builds a production-based asset-pricing model in which suppliers are matched with heterogeneous customers, choose capital and trade credit, face default/departure risk, and pay stochastic search costs after losing customers.
- **Quantitative result.** The calibrated model matches the key asset-pricing and network facts without targeting the counterparty premium directly. The model-implied spread is about **5.8% annually**, inside the empirical confidence interval.
- **Takeaway.** Trade credit contains information about both micro-level customer quality and macro-level search-cost risk. It is not merely working-capital finance; it is a contractual mechanism that stabilizes valuable production-network relationships.

2 Data and measurement

2.1 Firm sample

- The asset-pricing sample uses CRSP/Compustat common stocks, share codes 10 or 11, listed on NYSE, AMEX, or NASDAQ.
- Financial firms and utilities are excluded.
- The main sample spans **1978–2020**, because trade receivables data are sparse before 1978.
- Accounting variables are lagged before portfolio formation: portfolios are formed each June using the fiscal-year value from calendar year $t-1$, then held from July to the next June.

Interpretation: The June timing and accounting lag make the trading strategies implementable and reduce look-ahead bias.

2.2 Trade credit measure

The paper measures trade credit provision by supplier i in year t as

$$R/S_{i,t} = \frac{\text{Trade receivables}_{i,t}}{\text{Sales}_{i,t}}. \tag{1}$$

- Trade receivables are Compustat item RECTR.
- Sales are Compustat item SALE.
- $365 \times R/S$ is the accounting measure often called "days receivable."
- Observations with zero R/Sare excluded because such firms almost never extend trade credit.

Interpretation: A high R/Smeans the supplier delays collecting payment from customers. Economically, this can be viewed as providing liquidity insurance or selling at a discount through delayed payment.

2.3 Portfolio formation

- Low R/Sportfolio: firms at or below the 10th percentile of R/S.
- High R/Sportfolio: firms above the 90th percentile of R/S.
- Medium portfolio: firms between the two cutoffs.
- Portfolios are value weighted.
- The low and high portfolios are broad, with roughly 330 firms each.
- The low-minus-high return spread is called the **counterparty premium**.

Interpretation: The sort uses extreme deciles to sharpen the economic contrast between firms that extend little credit and firms that extend a lot.

2.4 Production-network data

- Supplier-customer links come from FactSet Revere.
- FactSet provides start and end dates for supplier-customer links, allowing link-duration measurement.
- The network sample covers much of the modern period and is used for firm-level duration and link-breaking tests.
- The authors clean and link FactSet to CRSP/Compustat using the procedure of Gofman, Segal, and Wu.

Interpretation: The network data are essential because the paper's mechanism is about customer retention, link duration, and search for new customers.

2.5 Aggregate search-cost proxies

The paper uses aggregate variables that should proxy for the difficulty of finding a new customer.

- **DeathMinusBirth**: predictable component of future establishment deaths minus births. A higher value means fewer potential new firms/customers, so search should be harder.
- **RelativeCompetition**: customer concentration minus supplier concentration. A higher value means customers have more bargaining power relative to suppliers, making matching more costly for suppliers.
- **Aggregate R/S**: the cross-sectional average receivables-to-sales ratio. The authors argue it is an observable proxy for aggregate customer-search costs.

Interpretation: Prediction 1 is that aggregate R/Sshould rise when customer-search costs rise, because firms extend more trade credit to keep existing customers.

2.6 Macroeconomic risk factors

In the pricing tests, the authors compare aggregate R/Sinnovations with other macro shocks:

- utilization-adjusted TFP innovations;
- corporate default spread;
- macroeconomic uncertainty;
- investment-specific technology shocks;
- intermediary capital;
- equity cost issuance shocks.

Interpretation: The point is not simply that a two-factor model fits better. The paper asks whether the specific aggregate object implied by the search-cost mechanism is the one that prices the R/Sportfolios.

3 Empirical specifications and identification logic

3.1 Portfolio return tests

The basic return object is the monthly excess return of portfolios sorted on lagged R/S. The key comparison is

$$r_t^{L-H} = r_t^{\text{Low } R/S} - r_t^{\text{High } R/S}. \tag{1}$$

- A positive average of r_t^{L-H} means low-R/Sfirms earn higher returns than high-R/Sfirms.
- Factor regressions project r_t^{L-H} on standard asset-pricing factors.
- The abnormal return α measures whether the spread is explained by existing factors.

Interpretation: The counterparty premium is economically large and not spanned by standard factor models, motivating a new risk-based mechanism.

3.2 Counterparty risk factor price

The paper tests whether a counterparty-risk factor is priced in cross-section. Conceptually, a stochastic discount factor is written as

$$M_t = 1 - b_f' f_t, \tag{2}$$

where f_t contains standard factors and the counterparty-risk factor.

Interpretation: The negative market price of counterparty risk means assets that covary negatively with counterparty-risk shocks require higher returns.

3.3 Link-duration regressions

The paper estimates annual Fama–MacBeth regressions of future link outcomes on supplier characteristics:

$$D_{s,t} = \text{const} + \beta_t' X_{s,t} + \varepsilon_{s,t}, \qquad t \in \{2003, \dots, 2020\}. \tag{4}$$

- $D_{s,t}$ is either future average link duration or a break indicator.
- $X_{s,t}$ includes industry-adjusted R/Sand controls: number of customers, investment, profitability, market value, book-to-market, and lagged duration.
- Predictors are standardized by unconditional standard deviations.

Interpretation: The key identification idea is within-year cross-sectional prediction: among comparable suppliers, do those extending more credit keep customers longer?

3.4 Aggregate trade credit and search costs

To test whether aggregate R/Sproxies for search costs, the paper estimates

$$\overline{R/S}_t = \rho_0 + \rho_{SC}\text{SearchCost}_t + \rho_{TFP} TFP_t + \epsilon_t. \tag{5}$$

- SearchCost is either DeathMinusBirth or RelativeCompetition.
- TFP controls for standard business-cycle fluctuations.
- A positive ρ_{SC} supports the idea that aggregate trade credit rises when search is harder.

Interpretation: This is a mechanism test: aggregate trade credit is not just a financial ratio, but a proxy for the shadow cost of customer replacement.

3.5 Macro SDF and pricing tests

The empirical two-factor SDF is

$$M_t = 1 - b_{MKTRF} MKTRF_t - b_{MACRO} MACRO_t. \tag{6}$$

The Euler condition implies

$$E[r_{i,t}^e] = \alpha_i + b_{MKTRF}\,\text{Cov}(r_{i,t}^e, MKTRF_t) + b_{MACRO}\,\text{Cov}(r_{i,t}^e, MACRO_t). \tag{7}$$

- Test assets are ten R/S-sorted portfolios.
- $MACRO_t$ is alternatively aggregate R/Sinnovations, TFP, default spread, uncertainty, IST, intermediary capital, or equity issuance-cost shocks.
- The paper asks which macro factor can make the pricing error of the low-minus-high R/Sspread insignificant.

Interpretation: Aggregate R/Sinnovations are the only macro variable in the horse race that explains the counterparty premium.

3.6 Cash-flow sensitivity test

The paper measures portfolio cash-flow exposure to aggregate R/Sinnovations as

$$CF_{i,t} = \beta_0 + \beta_1 MKTRF_t + \beta_2 \overline{R/S}_t + \epsilon_{i,t}. \tag{8}$$

- Cash flows are dividend growth, dividends per share, or EBIT/assets.
- Low-R/Sfirms have more negative cash-flow sensitivity to aggregate R/Sshocks.
- The low-minus-high cash-flow exposure is negative and significant.

Interpretation: This supports the mechanism because the same aggregate trade-credit shocks that explain returns also affect cash flows.

4 Model structure and economic mechanism

4.1 Production with customer quality

Supplier i produces

$$Y_{i,t} = (A_t C_{i,t})^{1-\alpha} K_{i,t}^{\alpha}, \tag{10}$$

where A_t is aggregate productivity and $C_{i,t}$ is the quality of the current customer.

Interpretation: A customer is not passive. Customer quality affects supplier output through productivity spillovers, fit, or markup opportunities in specialized production.

4.2 Aggregate productivity

Aggregate productivity follows

$$\log A_{t+1} = \log A_t + \mu_a + \sigma_a \varepsilon_{t+1}^{a}. \tag{11}$$

Interpretation: The model has standard aggregate productivity risk, but the central new risk is customer-search-cost risk.

4.3 Capital accumulation and adjustment costs

Capital evolves as

$$K_{i,t+1} = (1-\delta)K_{i,t} + I_{i,t}, \tag{12}$$

with adjustment cost

$$\phi(I_{i,t}, K_{i,t}) = b\left(\frac{I_{i,t}}{K_{i,t}} - \delta\right)^2. \tag{13}$$

Interpretation: Capital dynamics are conventional; the novelty is that relationship capital with customers also matters.

4.4 Trade credit and link termination

The customer link terminates with probability

$$\tau(r_{i,t+1}) = (\bar{p} - \underline{p})(1 - r_{i,t+1})^{\lambda} + \underline{p} \cdot g(a_t), \tag{14}$$

where $r_{i,t+1}$ is trade credit scaled by sales, and $g'(a_t) < 0$.

- More trade credit lowers idiosyncratic termination risk.
- The systematic component increases when aggregate productivity is low.
- The baseline uses $g(a) = \exp(a)^{\nu}$ with $\nu \leq 0$.

Interpretation: Trade credit is useful because it reduces customer departure/default risk. It is costly because it creates the possibility of unpaid receivables in bad states.

4.5 Customer replacement and search costs

If the customer departs, the supplier draws a new customer quality from a distribution. Search cost is

$$f_{t+1} = f_0 + \sigma_f \varepsilon^f_{t+1}. \tag{17}$$

- Search costs scale with aggregate productivity for stationarity.
- ε^f is a systematic counterparty-search-cost shock, orthogonal to productivity.
- This shock is the model's core counterparty-risk state.

Interpretation: Low-R/Sfirms are more likely to be forced into this search technology. Therefore they are more exposed to systematic fluctuations in search costs.

4.6 Stochastic discount factor

The model SDF is

$$M_{t,t+1} = \frac{\beta \exp\left(-\gamma_{a,t}\sigma_a\varepsilon^a_{t+1} - SGN \cdot \gamma_f \sigma_f \varepsilon^f_{t+1}\right)}{E_t\left[\exp\left(-\gamma_{a,t}\sigma_a\varepsilon^a_{t+1} - SGN \cdot \gamma_f \sigma_f \varepsilon^f_{t+1}\right)\right]}. \tag{18}$$

The aggregate-productivity price of risk is

$$\gamma_{a,t} = \exp(\gamma_a a_t). \tag{19}$$

Interpretation: Counterparty search shocks have a negative price of risk when they raise marginal utility. This is what makes exposure to search costs command a premium.

4.7 Cash flows and dynamic problem

Immediate net sales proceeds are

$$\widehat{D}_{i,t} = Y_t(1 - r_{i,t+1}) - \xi K_{i,t} - I_{i,t} - \phi(I_{i,t}, K_{i,t})K_{i,t}. \tag{20}$$

If the previous customer defaults, dividends are

$$D_{i,t} = \widehat{D}_{i,t} - f_{t-1}A_{t-1}. \tag{21}$$

If the previous customer does not default,

$$D_{i,t} = \widehat{D}_{i,t} + Y_{i,t-1}r_{i,t}. \tag{22}$$

Interpretation: The supplier chooses investment and trade credit to maximize market value. Trade credit shifts cash flows across periods and changes the probability of losing the customer.

5 Identification, interpretation, and credibility

5.1 What is identified empirically

- The paper does not claim random assignment of trade credit.
- It identifies a robust cross-sectional relation between R/S, expected returns, and customer-link duration.
- It then uses multiple mechanism tests to show that the relation aligns with costly customer search, not standard factor exposures or simple default risk.

Interpretation: The empirical strategy is triangulation: return spreads, factor tests, link-duration data, aggregate search-cost proxies, cash-flow exposures, and model calibration all point to the same mechanism.

5.2 Why the sign of the premium matters

- If trade credit were mainly about default risk, high-R/Sfirms should be riskier because they are exposed to unpaid receivables.
- The data show the opposite: high-R/Sfirms earn lower returns.
- This suggests high-R/Sfirms are safer because trade credit helps retain better customers and avoid costly customer search.

Interpretation: The sign of the premium rules out a simple "more receivables equals more default exposure" explanation.

5.3 Why standard factor models are not enough

- CAPM, FF3, FF4, FF5, and q-factor models leave large alphas.
- The low-minus-high R/Sspread alpha remains economically and statistically significant.
- The counterparty-risk factor is priced negatively across broader test assets.

Interpretation: Counterparty risk is not obviously subsumed by size, value, momentum, profitability, investment, or expected growth.

5.4 Why network evidence is central

- The paper observes actual supplier-customer link duration from FactSet.
- High-R/Sfirms have longer link durations and lower break probabilities.
- Firms that recently replace customers have lower dividends, profit margins, ROA, and sales growth.

Interpretation: This directly connects trade credit to the durability and cost of network relationships.

5.5 What the calibrated model adds

- The model checks whether the mechanism is quantitatively plausible.
- Calibration targets aggregate and firm moments, average R/S, link duration, and probability of breaking links.
- It does *not* target the counterparty premium directly.
- The model still generates a positive low-minus-high R/Sspread close to the data.

Interpretation: The model converts the qualitative mechanism into a quantitative asset-pricing explanation.

5.6 Limits and caveats

- R/Sis an endogenous choice; the empirical tests are not randomized treatment effects.
- FactSet coverage begins later than the return sample, so network tests use a shorter period.
- Some mechanisms are formalized in reduced form, including customer quality, liquidity shocks, and matching costs.
- The model abstracts from banks, trade-credit interest rates, multiple customer portfolios, and richer bargaining arrangements in the baseline.
- Internet Appendix checks address several of these issues, but the main paper's strongest claim is mechanism plausibility, not a single clean natural experiment.

Interpretation: The paper's credibility comes from consistency across many pieces of evidence rather than from one quasi-experimental shock.

6 Section-by-section roadmap

6.1 Introduction

- Establishes the puzzle: trade credit is large and central in production networks, but its asset-pricing implications are understudied.
- States the two novel facts: high R/Sfirms have lower risk premiums and longer customer relationships.
- Introduces the counterparty-risk mechanism: trade credit helps retain good customers and hedges costly search.
- Previews the production-based model and six predictions.

6.2 Section 1: Empirical facts

- Describes the CRSP/Compustat and FactSet Revere data.
- Defines R/Sas trade receivables over sales.
- Forms low, medium, and high R/Sportfolios.
- Documents the counterparty premium and shows it survives standard factor models.
- Shows high R/Sis associated with longer supplier-customer links and lower break risk.

6.3 Section 2: Explaining the facts in the data

- Develops six predictions from the costly-search hypothesis.
- Shows aggregate R/Srises with proxies for customer-search costs.
- Shows aggregate R/Sinnovations explain the counterparty premium.
- Shows customer search is associated with lower profitability.
- Shows firms provide more credit to better-quality customers.
- Shows short-duration supplier-customer links carry higher expected returns.

6.4 Section 3: Model

- Embeds trade credit in a production model with customer quality.
- Suppliers choose capital and trade credit.
- Trade credit lowers the probability of customer departure/default.
- Losing customers triggers costly search for new counterparties.
- Search-cost shocks are systematic and negatively priced.

6.5 Section 4: Explaining the facts in theory

- Calibrates model parameters to aggregate, firm-level, and link-duration moments.
- Shows the model matches the low-minus-high R/Sspread and link-duration pattern.
- Demonstrates that the model-implied aggregate R/Sshock prices the R/Sspread.
- Inspects policy functions and risk decompositions.
- Performs sensitivity analysis around pricing of search shocks, average search costs, systematic defaults, and capital adjustment costs.

6.6 Section 5: Conclusion

- Summarizes that trade credit reveals customer quality and search-cost risk.
- Positions aggregate R/Sas a macro proxy for costly customer search.
- Suggests financial-network analogues, such as credit lines and relationship duration among banks.

7 Figures and tables

7.1 Tables 1–2: the counterparty premium and factor-model alphas

Read:

- Table 1 shows a monotonic return decline from low to high R/Sfirms.
- Low R/Sfirms earn 1.253% monthly, high R/Sfirms earn 0.617% monthly, and the low-minus-high spread is 0.636% monthly.
- Industry-adjusted R/Sgives a similar spread of 0.605% monthly.
- Table 2 shows the spread has large alphas under CAPM, FF3, FF4, FF5, and q5 models.
- The CAPM alpha is 0.863% monthly; even the q5 alpha is 0.533% monthly.

Economic message: Low trade-credit suppliers earn a large premium that is not explained by standard risk factors.

Table 1
Portfolios sorted on R/S

Portfolio	A. R/S Mean	A. R/S SD	B. R/S^{IA} Mean	B. R/S^{IA} SD
Low R/S	1.253	4.868	1.349	5.479
Medium	1.114	4.590	1.096	4.533
High R/S	0.617	5.987	0.744	5.868
Spread	0.636	4.014	0.605	3.720
(L-H)	(3.28)		(4.01)	

The table reports the average monthly returns of portfolios sorted on the trade receivables to sales (R/S) ratio and the spread between the returns of the low and high R/S portfolios. The low (high) R/S portfolio includes all firms with R/S ratios below (above) the 10th (90th) percentile of the cross-sectional distribution of R/S ratios from fiscal year ending in calendar year $t-1$. Panel A reports the returns of portfolios constructed using the R/S ratio defined in Equation (1), whereas panel B reports the returns of portfolios constructed using the industry-adjusted R/S ratio, denoted by R/S^{IA}. This industry adjustment at time t is implemented by (a) assigning each firm to its Fama-French 30 industry group and (b) subtracting each industry's time-t cross-sectional median R/S ratio from each firm's R/S ratio. Both panels report value-weighted returns. Mean refers to the average monthly return and SD denotes the standard deviation of monthly returns. Parentheses report Newey and West (1987) robust t-statistics. All portfolios are formed at the end of each June from 1978 to 2020 and are rebalanced annually. Thus, portfolio returns span July 1978 to December 2020.

(a) Table 1: portfolios sorted on R/S.

Table 2
Value-weighted R/S spread and factor models

	CAPM	FF3F	FF4F	FF5F	q^5
MKTRF	−0.320	−0.285	−0.261	−0.244	−0.233
	(−6.29)	(−5.49)	(−4.86)	(−4.92)	(−4.22)
SMB		−0.100	−0.104	0.011	0.034
		(−1.35)	(−1.36)	(0.14)	(0.46)
HML		0.119	0.169	0.040	
		(0.12)	(0.17)	(0.45)	
MOM			0.118		
			(2.02)		
Profit.				0.399	0.346
				(3.81)	(2.72)
Invest.				0.103	0.254
				(0.90)	(2.61)
EG					0.145
					(0.15)
α	0.863	0.833	0.741	0.643	0.533
	(4.68)	(4.65)	(4.24)	(3.76)	(3.19)

The table reports the results of time-series regressions of the value-weighted counterparty premium on common risk factors. Parameter estimates are obtained by regressing monthly excess returns on each set of monthly risk factors. MKTRF is the excess return of the market portfolio. SMB and HML are the size and value factors of the Fama and French (1993), whereas MOM is the momentum factor of Carhart (1997). Profit. and Invest. correspond to the RMW and CMA factors (ROE and I/A factors) of the Fama and French (2015) five-factor (Hou et al. (2021) q^5-factor) model, while EG represents the expected growth factor from the q^5 model. Values in parentheses report Newey and West (1987) robust t-statistics. Returns span July 1978 to December 2020.

(b) Table 2: factor-model alphas.

7.2 Tables 3–4: pricing of counterparty risk and network characteristics

Read:

- Table 3 estimates negative prices of counterparty risk across 25, 74, and 154 test portfolios.
- Adding the counterparty risk factor reduces mean absolute pricing errors.
- Table 4 shows low and high R/Sfirms differ strongly in upstreamness and duration, but not in customer concentration in a way that would simply imply bargaining-power differences.
- High R/Sfirms have longer customer-link duration: 48.69 months versus 40.88 months for low R/Sfirms.

Economic message: The counterparty-risk factor is priced, and the network data show that R/Sis linked to relationship duration.

Table 3
The market price of counterparty risk

A. CAPM plus the counterparty risk factor

	25 portfolios		74 portfolios		154 portfolios	
	CAPM	CPR	CAPM	CPR	CAPM	CPR
b_{MKTRF}	3.662	8.369	3.557	5.260	3.376	6.181
	(3.36)	(4.74)	(3.35)	(4.62)	(3.35)	(6.14)
b_{CPR}		−14.167		−5.199		−8.360
		(−3.35)		(−3.57)		(−7.09)
MAE	0.706	0.585	0.886	0.793	0.923	0.744

B. FF3F plus the counterparty risk factor

	25 portfolios		74 portfolios		154 portfolios	
	FF3F	CPR	FF3F	CPR	FF3F	CPR
b_{MKTRF}	3.763	9.352	4.253	5.408	4.436	6.161
	(3.24)	(4.83)	(3.91)	(4.67)	(4.40)	(5.98)
b_{SMB}	0.542	2.874	−2.384	−1.263	−3.796	−1.940
	(0.34)	(1.50)	(−1.51)	(−0.79)	(−2.61)	(−1.31)
b_{HML}	3.622	0.539	0.318	−0.483	1.087	−0.200
	(2.11)	(0.25)	(0.19)	(−0.29)	(0.67)	(−0.12)
b_{CPR}		−20.147		−4.693		−6.940
		(−4.01)		(−3.16)		(−5.54)
MAE	0.617	0.505	0.879	0.801	0.804	0.717

The table reports the estimates of the risk factor loadings associated with both the CAPM (in panel A) and the Fama and French (1993) three-factor model (in panel B) when each of these models is estimated with and without the counterparty risk factor. Here, the counterparty risk factor (CPR) is constructed by buying firms with high R/S ratios and selling firms with low R/S ratios. All firms underlying each R/S portfolio are value weighted. Each model is estimated by generalized methods of moments (GMM) using the moment conditions $\mathbb{E}\left[M_t r^e_{i,t}\right]=0$, where $r^e_{i,t}$ represents the excess return of test asset i at time t and M_t denotes the stochastic discount factor. We assume that M_t is specified as $M_t = 1 - \boldsymbol{b}'\boldsymbol{f}_t - b_{CPR}CPR_t$, where $\boldsymbol{f}_t$ represents the common factors associated

(a) Table 3: market price of counterparty risk.

Table 4
Network-related characteristics of the R/S portfolios

	Low (L)	Medium	High (H)	Diff(L-H)	t(Diff)
Centrality	0.30	0.55	0.50	−0.20	(−1.27)
Upstreamness	1.76	2.73	3.06	−1.30	(−11.92)
HHI (Customer)	0.04	0.05	0.05	−0.00	(−0.42)
IVOL (Customer)	1.73	1.51	1.50	0.23	(1.12)
Loan spread (Customer)	1.84	1.90	1.87	−0.03	(−0.08)
Secured debt (Customer)	0.43	0.39	0.30	0.13	(0.92)
Debt covenants (Customer)	0.60	0.55	0.41	0.18	(1.44)
Duration	40.88	47.18	48.69	−7.81	(−3.19)

The table shows the value-weighted network-related characteristics of the portfolios sorted on the trade receivables to sales (R/S) ratio. The sample period underlying this table spans June 2003 to 2020, as the FactSet Revere database is only available beginning in April 2003. Section IA.1 of the Internet Appendix provides details about the construction of each variable. The column Diff(L-H) refers to the difference between the average characteristics of the low and high R/S portfolios, and t(Diff) is the Newey and West (1987) t-statistic associated with this difference.

(b) Table 4: network characteristics.

7.3 Tables 5–6: link duration and aggregate search costs

Read:

- Table 5 shows a one-standard-deviation increase in R/Spredicts about four more months of future link duration.
- The same increase in R/Spredicts a lower link-break probability, with coefficient about -0.09.
- Table 6 shows aggregate R/Sis positively related to search-cost proxies: DeathMinusBirth and RelativeCompetition.
- TFP alone does not explain aggregate R/S, but search-cost measures have strong positive correlations after TFP controls.

Economic message: Trade credit is linked to longer relationships at the firm level and to higher search costs at the macro level.

Table 5
Predicting the length of supplier-customer links

	A. Future duration		*B. Pr(Break=1)*	
Constant	44.63	44.57	0.63	0.62
	(6.90)	(6.91)	(20.79)	(21.64)
R/S	4.15	4.09	−0.09	−0.09
	(3.01)	(3.07)	(−4.39)	(−4.15)
ln(ME)	0.14	0.35	0.03	0.04
	(0.44)	(0.91)	(4.94)	(6.27)
ln(B/M)	−0.12	−0.08	0.01	0.01
	(−0.35)	(−0.25)	(0.44)	(0.60)
I/K	−0.32	−0.35	−0.00	−0.00
	(−0.48)	(−0.52)	(−0.40)	(−0.31)
ROA	1.25	1.22	−0.02	−0.02
	(3.92)	(3.92)	(−3.59)	(−3.91)
Lagged duration	5.13	5.14	−0.09	−0.09
	(5.49)	(5.45)	(−6.65)	(−5.91)
Number of customers		−0.66		−0.02
		(−2.44)		(−2.39)
R^2 (%)	14.25	14.26	10.18	11.22

The table reports the results of Fama-MacBeth regressions that use industry-adjusted supplier-level characteristics to predict the average duration, D_t, of each supplier's link with its customers, measured in months (panel A) and the probability that a supplier-customer link breaks (panel B). The regression specification is presented in Equation (4) and the procedure we follow to estimate these regressions is described in Section 1.3.2. Our measure of link duration is either (1) the average future duration of each supplier's links with its customers (measured in months in panel A) or (2) an indicator variable Break that is equal to one in the situation in which at least half or more of a supplier's customers at time t are no longer active customers in 4 years' time. Finally, Newey and West (1987) robust t-statistics are reported in parentheses and each supplier-level characteristic is standardized by dividing the characteristic by its unconditional standard deviation. The table also reports the maximal share of D_t explained by the supplier-level characteristics. The time period for the analysis ranges from June 2003 to June 2020.

(a) Table 5: predicting link duration and breaks.

Table 6
Aggregate trade credit extension and customer search costs

	(1)	(2)	(3)	(4)	(5)
DeathMinusBirths		0.37	0.43		
		(3.53)	(4.42)		
RelativeCompetition				0.56	0.54
				(3.41)	(3.35)
TFP	0.24		0.38		0.19
	(1.13)		(2.48)		(1.13)
$\bar{R}^2$ (%)	3.27	10.20	21.61	29.63	31.46

The table reports the results of the projection $\overline{R/S}_t = \rho_0 + \rho_{SearchCost} SearchCost_t + \rho_{TFP} TFP_t + \varepsilon_t$, where $\overline{R/S}_t$ is the aggregate receivables-to-sales ratio at time t, $SearchCost_t$ is a macroeconomic variable that is associated with an increase in customer-search costs, and TFP_t is the utilization-adjusted total factor productivity from Fernald (2012). The first measure of customer search costs is "DeathMinusBirth," which is the expected difference between the death and birth rate of establishments from the BLS. This expected death-minus-birth rate of establishments is obtained from the fitted value of a regression of the death-minus-birth rate at time $t+1$ on a constant, the current death-minus-birth rate of establishments at time t, and other time-t predictors including the corporate credit spread and the excess returns of the market portfolio. The second measure of customer search costs, "RelativeCompetition," captures the concentration of customers relative to suppliers, and is constructed in three steps. First, in each quarter, each firm in the sample is denoted a customer (supplier) firm if its upstreamness score from Gofman, Segal, and Wu (2020) is below (above) the 30th (70th) percentile for that period. Next, the Herfindahl-Hirschman index (HHI) of the group of customers and suppliers is computed. Finally, the HHI of the supplier firms is subtracted from the HHI of the customer firms. In all regressions both the dependent and independent variables are standardized, and Newey and West (1987) t-statistics are reported in parentheses. The sample period is annual from 1978 (1992) to 2020 for regressions including RelativeCompetition (DeathMinusBirths).

(b) Table 6: aggregate trade credit and search costs.

7.4 Table 7 and Figure 1: aggregate R/Sinnovations as search-cost shocks

Read:

- Table 7 shows that the CAPM leaves a 9.97% annual pricing error for the R/Sspread.
- Using aggregate R/Sinnovations as the macro factor reduces the spread pricing error to -0.84%, statistically indistinguishable from zero.
- Other macro shocks do not remove the pricing error.
- The R/Sspread return covaries negatively with aggregate R/Sinnovations, with covariance -0.77.
- Panel C estimates a negative market price of R/Srisk: $b_{R/S} = -14.50$ using R/Sportfolios and -6.91 using broader test assets.
- Figure 1 plots aggregate R/Sand its innovations, visually motivating the macro factor.

Economic message: The same aggregate object implied by the search-cost mechanism explains the cross-sectional premium.

7.5 Tables 8–9: customer replacement costs and duration premium

Read:

- Table 8 shows firms recently searching for new customers have lower dividends per share, lower profit margins, lower ROA, and lower sales growth.
- These differences support the assumption that customer search and replacement are costly.
- Table 9 shows firms with low duration links earn higher monthly returns.
- The low-minus-high duration spread is 0.972% monthly in the raw duration sort and 0.490% monthly in the industry-adjusted duration sort.
- The duration spread is also related to aggregate R/Srisk exposures.

Table 7
Pricing R/S-sorted portfolios using macroeconomic variables

A. Model-implied alphas

	MKTRF	$\overline{R/S}$	TFP	DEF	UNC	IST	INT	ECS
Low R/S (L)	9.08	−0.95	9.08	8.40	6.51	−2.57	6.73	8.59
Medium R/S	−4.03	1.55	5.34	8.84	4.94	−1.20	4.77	5.44
High R/S (H)	−11.53	−0.11	1.63	2.84	−1.36	−0.53	0.62	−2.19
Spread	9.97	−0.84	7.44	5.55	7.87	−2.04	6.11	10.78
(L-H)	(4.02)	(−0.37)	(3.18)	(2.11)	(2.22)	(−1.97)	(2.04)	(3.96)

B. Covariance with macroeconomic variable

	$\overline{R/S}$	TFP	DEF	UNC	IST	INT	ECS
Low R/S (L)	−0.65	−3.12	−0.88	0.12	0.26	−2.93	26.47
Medium R/S	−0.31	−5.75	−2.18	−0.01	0.20	−5.01	51.56
High R/S (H)	0.12	−0.38	−3.12	−0.25	0.06	−17.71	27.54
Spread	−0.77	−2.75	2.24	0.37	0.20	14.79	−1.07
(L-H)	(−2.65)	(−0.66)	(1.59)	(0.81)	(3.01)	(1.18)	(−0.04)

C. Market prices of risk

	R/S portfolios		Broader test assets	
b_{MKTRF}	4.38	1.59	4.69	1.41
	(4.47)	(1.90)	(14.20)	(4.41)
$b_{\overline{R/S}}$		−14.50		−6.91
		(−4.70)		(−9.21)
MAE	5.60	1.35	7.82	1.09

D. Cash flow sensitivity

	Div. growth		Div. per share		EBIT / AT	
Portfolio	β	$t(\beta)$	β	$t(\beta)$	β	$t(\beta)$
Low R/S (L)	−0.34	(−5.38)	−11.17	(−9.05)	−0.06	(−0.47)
Medium R/S	−0.29	(−4.04)	−9.06	(−6.16)	0.51	(3.95)
High R/S (H)	−0.01	(−0.17)	1.98	(0.89)	0.29	(2.20)
Spread (L-H)	−0.33	(−4.94)	−13.16	(−5.09)	−0.35	(−2.70)

The table reports statistics related to the ability of different macroeconomic shocks to price the R/S-sorted portfolios. We posit that the SDF is given by Equation (6) and is driven by the market's excess return ($MKTRF_t$) and an additional macroeconomic shock ($MACRO_t$), including (a) innovations to the aggregate receivables-to-sales ratio ($\overline{R/S}$), measured by its log first difference, (b) innovations to the utilization-adjusted TFP from Fernald (2012) (TFP), measured by its log first difference, (c) the corporate default spread (DEF), (d) macroeconomic uncertainty from Jurado, Ludvigson, and Ng (2015) (UNC), (e) investment-specific technology shocks proposed by Papanikolaou (2011) (IST), (f) intermediary capital from He, Kelly, and Manela (2017), and (g) the equity cost issuance shock from Belo, Lin, and Yang (2019) (ECS). We use GMM to estimate Equation (7) using the 10 R/S-sorted portfolios, constructed following the procedure in Section 1.2, as test assets. Panel A reports model-implied pricing errors (α) associated with the GMM procedure, and panel B reports the covariance between portfolio returns and the macroeconomic shocks $MACRO_t$. The leftmost column in panel A, entitled $MKTRF$, omit $MACRO_t$ from the SDF, such that the SDF includes only the market's excess returns. Panels C and D focus on aggregate trade credit as the macroeconomic variable of interest. Specifically, panel C reports the market price of risk associated with $MKTRF_t$ and $\overline{R/S}$ innovations when the test assets are either (a) the 10 R/S-sorted portfolios or (b) a broader set of test assets that features the five Fama-French industry portfolios, six portfolios sorted on size and book-to-market, six portfolios sorted on size and profitability, and six portfolios sorted on size and investment. Finally, panel D reports the sensitivity of the cash flows underlying the R/S-sorted portfolios to variations in aggregate trade credit. The sensitivity of each portfolio corresponds to β_2 from Equation (8). Cash flows are measured using (a) divided growth, which is constructed by compounding the difference between the total returns and the returns excluding dividends of each portfolio, (b) the average dividends per share of each portfolio, or (c) the average profitability of each portfolio, measured using EBIT-to-assets. Each of these cash flow measures are industry-adjusted. The data underlying this analysis are annual and spans 1978 to 2020. Newey and West (1987) t-statistics are reported in parentheses.

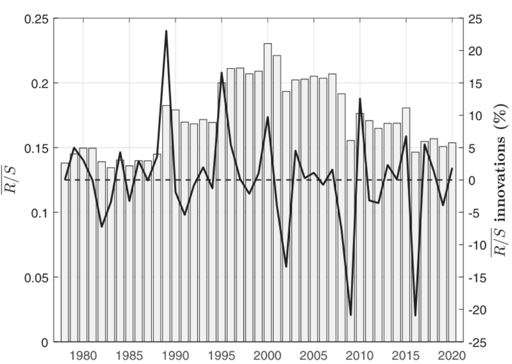

Figure 1
Innovations to the aggregate receivables-to-sales ratio
The figure shows the time series of the aggregate receivables-to-sales ratio $\overline{R/S}$ (gray bars) and its innovations (solid line). Innovations are computed as the logarithm of changes in the aggregate receivables-to-sales ratio. The sample spans from 1978 to 2020.

Economic message: Search is costly in cash-flow terms, and short-duration links command higher returns.

Table 8
Searching for new customers and profit margins

Characteristic	$\mathbb{I}_{NewCustomers}=0$	$\mathbb{I}_{NewCustomers}=1$	Difference	t(Diff.)
Div. per share	0.26	0.17	0.09	(10.70)
Profit margin	0.06	−0.22	0.27	(6.80)
ROA (%)	1.72	0.98	0.74	(13.88)
Operating costs	−0.02	0.01	−0.03	(−8.74)
TFP	0.15	0.05	0.10	(9.65)

The table reports the value-weighted characteristics of suppliers that recently searched for and matched with most of its customers (i.e., most of the customers are new), denoted by $\mathbb{I}_{NewCustomers}=1$, and suppliers that did not recently search for new customers, denoted by $\mathbb{I}_{NewCustomers}=0$. For each firm i and each quarter t between June 2003 and December 2020 (i.e., for each quarter for which the FactSet data are available), we set the indicator $\mathbb{I}_{NewCustomers}$ equal to one if at least 50% of the links between firm i and its customers in quarter t did not exist at the end of quarter $t-1$. We then compute the average profitability, operating costs, and idiosyncratic productivity across all firms assigned to each group. We measure profitability using the quarterly values of dividends per share, profit margin, and ROA, we measure operating costs using the quarterly measure of Novy-Marx (2011), and we measure firm-level productivity following Imrohoroglu and Tuzel (2014). The column denoted by Difference reports the difference between the average characteristics for which $\mathbb{I}_{NewCustomers}=0$ and for which $\mathbb{I}_{NewCustomers}=1$, and the column denoted by t(Diff) is the Newey and West (1987) t-statistic associated with this difference.

(a) Table 8: customer search and profitability.

Table 9
Portfolios sorted on link duration: Returns, pricing errors, and risk exposures

	A. Duration portfolio returns		*B. Duration^{IA}* portfolio returns		*C. Pricing errors* implied by $\overline{R/S}$		*D. Covariance* with $\overline{R/S}$	
Portfolio	Mean	SD	Mean	SD	α	$t(\alpha)$	Cov.	t(Cov.)
Low DUR	2.011	4.625	1.494	4.831	0.49	(0.47)	−0.02	(-2.42)
Medium	1.038	4.060	1.054	4.050	0.74	(1.35)	−0.00	(-0.39)
High DUR	1.039	4.257	1.004	4.502	0.83	(1.39)	−0.01	(-1.12)
Spread	0.972	2.744	0.490	2.483	-0.34	(-0.55)	−0.02	(-2.52)
(L-H)	(4.61)		(3.05)					

The table reports the average monthly returns of link duration (DUR) sorted portfolios and their pricing when we estimate Equation (7) using innovations to aggregate trade credit as $MACRO_t$. At the end of each month from June 2003 onward, the cross-section of firms is sorted into three portfolios based on the $10th$ and $90th$ percentiles of the cross-sectional distribution of link duration at the end of the previous month. As such, portfolio returns span July 2003 to December 2020. Panel A reports the returns of portfolios constructed using link duration as a sorting variable, and panel B reports the returns of portfolios sorted on industry-adjusted link duration, denoted by Duration^{IA}. This industry adjustment is implemented by (a) assigning each firm to its Fama-French 30 industry group and (b) subtracting each industry's cross-sectional median link-duration from each firm's link duration. Here, Mean refers to the average value-weighted monthly return, and SD denotes the standard deviation of monthly returns. Panels C and D report the results of estimating the moment condition denoted by Equation (7) using GMM. Here, the macroeconomic shocks $MACRO_t$ are innovations to the aggregate receivables-to-sales ratio ($\overline{R/S}$), as measured by its log first difference. The test assets are the 10 duration-sorted portfolios that are constructed following the procedure outlined above. Panel C shows the model-implied pricing errors associated with the GMM procedure, and panel D shows the covariance of each portfolio returns with the macroeconomic shocks. The data underlying this GMM analysis is annual and spans 2003 to 2020. Parentheses in each panel report Newey and West (1987) robust t-statistics.

(b) Table 9: duration-sorted portfolios.

7.6 Tables 10–11: calibration and model fit

Read:

- Table 10 reports benchmark calibration: technology, capital, liquidity shock probabilities, and SDF parameters.
- The model is calibrated to aggregate and firm-level moments, link duration, and break probabilities.
- Table 11 compares data and model moments.
- The model matches the average R/S, dispersion of investment and sales growth, and output growth volatility.
- The model also matches the low/medium/high R/Sreturn pattern and link-duration pattern.

Economic message: The quantitative model is disciplined by moments other than the main return spread, then evaluated on its ability to reproduce the premium and link facts.

Table 10
Model calibration

Symbol	Value	Parameter	Symbol	Value	Parameter
A. Technology			*B. Capital*		
μ_a	2%	Productivity growth rate	δ	8%	Capital depreciation rate
σ_a	2.7%	Productivity shock volatility	α	0.4	Capital share of output
σ_c	0.6	Counterparty quality dispersion	b	0.9	Quadratic adjustment cost
f_0	0.49	Average matching cost	ξ	2.0	Fixed operating cost
σ_f	0.1	Matching cost volatility			
C. Liquidity shock probability			*D. SDF*		
$\overline{p}$	0.37	Pr(Shock if R/S=0)	β	0.979	Time discount factor
$\underline{p}$	0.24	Pr(Shock if R/S$\to\infty$)	γ_a	−84.1	Price of productivity shocks
λ	10	Convexity of shock function	γ_f	7.7	Magnitude of priced counterparty shocks
ν	−0.70	Cyclicality of liquidity shock	SGN	−1	Sign of priced counterparty shocks

The table shows the parameters of the benchmark model calibration. The model is calibrated at the annual frequency.

(a) Table 10: calibration.

Table 11
Model-implied moments against data

Statistic	Data	Model	Statistic	Data	Model
A. Aggregate moments			*B. Firm-level moments*		
Output growth:			*Receivables-to-sales (R/S):*		
Mean	2.51	1.97	Mean	19.50	19.48
Volatility	2	2.09	Volatility	5.55	4.58
Autocorrelation	0.23	0.32	Autocorrelation	0.45	0.42
Equity premium:			*Investment-to-capital (I/K):*		
Mean	8.78	8.67	Volatility	13.46	14.18
Volatility	15.98	15.82	Autocorrelation	0.33	0.24
Sharpe ratio	0.54	0.54	*Other:*		
Productivity and doubtful receivables growth:			Sales growth volatility	27.51	29.92
Correlation	−0.10	−0.09	Operating profits / sales volatility	10.80	10.63
C. R/S portfolio returns			*D. R/S portfolio duration*		
Low R/S	15.04	15.24	Low R/S	3.40	3.46
Medium R/S	13.37	11.75	Medium R/S	3.90	3.95
High R/S	7.40	9.40	High R/S	4.06	4.06
Spread	7.63	5.84	Spread	0.66	0.60

The table shows model-implied moments against their empirical counterparts. Panel A and B show moments related to aggregate and firm-level quantities, and panels C and D show the average returns and link durations of the R/S-sorted portfolios. All model-implied moments are based on model simulated data for 5,000 periods (years) and 1,000 firms. The model-implied sorting procedure in panels C and D is identical to the empirical strategy described in Section 1. That is, the low (high) R/S firms refers to the firms in the bottom (top) 10% of the cross-sectional distribution of the R/S ratio.

(b) Table 11: model-implied moments.

7.7 Table 12 and Figure 2: model-implied pricing and policy functions

Read:

- Table 12 reports model-implied prices of risk, R/Sreturn exposures, and cash-flow sensitivities.
- The model-implied $b_{R/S}$ is negative, and the spread alpha is small.
- Low R/Sportfolios have more negative exposure to aggregate R/Sshocks than high R/Sportfolios.
- Figure 2 shows the model policy for trade credit as a function of customer quality.
- Trade credit rises with customer quality: firms provide more credit to customers worth retaining.
- The right panel shows that higher trade credit lowers the customer's liquidity/default probability.

Economic message: The model's optimal policy connects micro customer quality to macro risk exposures and expected returns.

Table 12
Pricing R/S-sorted portfolios: Model-implied risk prices and exposures

A. Prices of risk and α			B. Risk exposures to $\overline{R/S}$			C. Cash flow sensitivity to $\Delta\overline{R/S}$		
b_{MKTRF}	5.57	[4.66, 7.14]	Low R/S (L)	−0.43	[−0.49, −0.37]	Low R/S (L)	−0.55	[−0.59, −0.49]
$b_{\overline{R/S}}$	−5.91	[−7.48, −4.29]	Medium R/S	−0.28	[−0.33, −0.23]	Medium R/S	−0.02	[−0.03, −0.02]
			High R/S (H)	−0.16	[−0.21, −0.12]	High R/S (H)	0.03	[0.00, 0.05]
α_{Spread}	0.21	[−0.07, 0.53]	Spread	−0.27	[−0.30, −0.24]	Spread	−0.57	[−0.61, −0.54]

The table shows the output implied by estimating the moment condition given in Equation (7) via GMM using model-simulated data. The macroeconomic shocks $MACRO_t$ are model-implied innovations to the aggregate receivables-to-sales ratio ($\overline{R/S}$), measured using the log first difference of the ratio. The test assets are 10 model-implied R/S-sorted portfolios, constructed using the same procedure used in the data. Panel A shows the model-implied market prices of risk for the market factor (b_{MKTRF}) and for innovations to the aggregate receivables-to-sales ratio ($b_{\overline{R/S}}$). The bottom row of the panel shows the pricing error (α) for the model-implied R/S-spread implied by this two-factor SDF. Panel B shows the exposure (covariance) of each model-implied R/S-sorted portfolio's returns with $\overline{R/S}$ innovations. Panel C reports the sensitivity of the cash flows of each R/S-sorted portfolio to $\overline{R/S}$. All moments are obtained from 100 finite-sample simulations of the economy, in which the analyses described in Section 2.2 are applied to the simulated data. Finally, values in brackets are the 90% confidence interval for each moment across the finite-sample simulations.

(a) Table 12: model pricing and exposures.

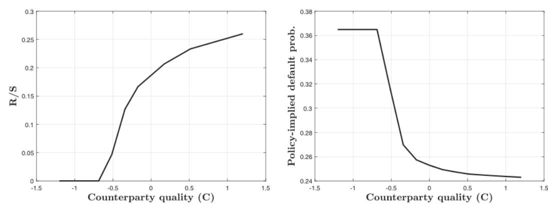

Figure 2
Model-implied policy functions
The left panel shows the model-implied policy for a firm's (supplier's) R/S (r) against different values of counterparty (customer) quality (C). The right panel shows the liquidity probability ($\Gamma(r)$ from Equation (14)) for the counterparty as implied by the R/S policy. Both policies are plotted when all other state variables are set to their stochastic steady-state values.

(b) Figure 2: policy functions.

7.8 Table 13: sensitivity analysis

Read:

- The benchmark model generates an R/Sspread of 5.84% annually versus 7.63% in the data.
- If counterparty search-cost shocks are not priced ($\gamma_f = 0$), the spread collapses to 0.07%.
- If counterparty shocks are positively priced, the spread becomes negative.
- Increasing average search costs raises the spread; reducing them lowers it.
- More systematic default risk reduces the spread because high-R/Sfirms become riskier through possible receivable losses.
- Higher capital adjustment costs raise the spread to 7.38%, closer to the data, but reduce investment volatility.

Economic message: The sensitivity analysis shows that negatively priced counterparty-search risk is the key source of the premium.

Table 13
Counterparty premium: Sensitivity analysis

Row	Moment	Mean(R/S)	$\sigma(I/K)$	$\sigma(\Delta$Sales$)$	$\sigma(\Delta$GDP$)$	ρ(TFP,Doubt R/S)	R/S spread
(1)	Data	19.50	13.46	27.51	2.00	−0.10	7.63
(2)	Benchmark [Calibration as in Table 10]	19.48	14.18	29.92	2.09	−0.09	5.84
(3)	$\gamma_f = 0$ [Counterparty shocks not priced]	19.30	13.35	30.02	2.09	−0.08	0.07
(4)	$SGN = 1$ [Counterparty shocks positively priced]	18.88	12.88	29.87	2.10	−0.07	−0.76
(5)	$f_0 \uparrow (f_0 = 0.50)$ [Higher avg. customer-search cost]	19.81	14.36	30.02	2.12	−0.09	6.61
(6)	$f_0 \downarrow (f_0 = 0.48)$ [Lower avg. customer-search cost]	19.20	14.33	30.11	2.11	−0.09	5.17
(7)	$\lvert\nu\rvert \uparrow (\nu = -1.2)$ [Customer default in worse states]	19.33	14.39	29.95	2.11	−0.15	4.39
(8)	$\lvert\nu\rvert \downarrow (\nu = 0.0)$ [Customer default idiosyncratic]	19.53	14.16	30.08	2.08	0.00	10.23
(9)	$b \uparrow (b = 2)$ [Higher capital adj. cost]	19.52	10.25	30.16	2.05	−0.08	7.38
(10)	$b \downarrow (b = 0.5)$ [Lower capital adj. cost]	19.29	17.74	30.14	2.17	−0.09	5.16

The table shows the model-implied average firm-level R/S ratio (Mean(R/S)), investment volatility ($\sigma(I/K)$), sales growth volatility ($\sigma(\Delta$Sales$)$), aggregate output growth volatility ($\sigma(\Delta$GDP$)$), the correlation between aggregate productivity and the growth rate of doubtful receivables (ρ(TFP,Doubtful R/S)), and the counterparty premium across alternative calibrations of the model. Each alternative calibration is contrasted with both (a) the empirical quantity and (b) the benchmark value of each moment in the first two rows of the table.

Table 13: counterparty premium sensitivity analysis.

8 Short summary

- Trade credit is measured as trade receivables over sales.
- Low R/Sfirms earn higher expected returns than high R/Sfirms.
- The low-minus-high spread is about 7.6% per year.
- Standard factor models do not explain this spread.
- High R/Sfirms maintain longer supplier-customer relationships.
- A one-standard-deviation increase in R/Spredicts roughly four more months of link duration and lower break probability.
- Aggregate R/Srises with proxies for customer-search costs.
- Aggregate R/Sinnovations explain the counterparty premium while many other macro shocks do not.
- Firms that recently replace customers exhibit lower profitability, suggesting customer search is costly.
- Firms extend more trade credit to higher-quality customers.
- Firms with shorter-duration links earn higher returns.
- The production model features supplier-customer matches, heterogeneous customer quality, costly search, customer default/departure risk, and trade credit as a retention mechanism.
- Trade credit lowers the probability of losing customers but exposes the supplier to unpaid receivable risk.
- Under the calibrated model, the search-cost hedging benefit dominates the default-risk cost for high R/Sfirms.
- The model produces a positive counterparty premium even though it does not directly target the premium.
- The mechanism is: low R/S$\Rightarrow$ worse customers and shorter links $\Rightarrow$ more customer replacement $\Rightarrow$ greater exposure to systematic search-cost shocks $\Rightarrow$ higher expected returns.

9 Conclusion

9.1 Core conclusion

Grigoris, Hu, and Segal show that trade credit contains asset-pricing information because it reveals how exposed suppliers are to counterparty search risk. Firms that extend little trade credit are more exposed to costly customer replacement and therefore earn higher expected returns. Firms that extend more trade credit keep better customers longer and are safer.

9.2 Economic intuition

Suppliers face two risks when dealing with customers. First, trade credit may not be repaid if customers default. Second, losing customers forces suppliers to search for new counterparties, and this search is costly especially in bad aggregate states. The paper argues that the second risk is more important for expected returns. Trade credit is valuable because it helps retain high-quality customers and reduces the chance that a supplier must pay systematic search costs.

9.3 Takeaway

The paper turns a balance-sheet working-capital variable into a measure of production-network risk. Receivables are not just short-term financing; they are a signal of relationship quality, customer-retention policy, and exposure to aggregate customer-search costs. This makes trade credit relevant for both asset prices and the dynamics of production networks.